

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

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I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Industry Problem Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed

at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost

implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the

industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design

improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination	Good collaboration	Limited participation	No collaboration or contribution

1. Startup Network Design using Star Topology:

Introduction:

A star topology is suitable for a 20-person startup because it is easy to manage, troubleshoot, and expand.

Devices Used:

- 24-port Layer 2 / Layer 3 managed switch
- Router / Firewall
- Wi-Fi Access Point
- Workstations, printers and VoIP Phones

Transmission Media:

Cat6 UTP cables

Ethernet RJ45 connectors.

Network Design:

- Each workstation is connected directly to the central switch
- The switch acts as the communication hub
- The router / firewall connects the local network to the internet.
- The access point provides wireless connectivity for mobile devices.

Working Principle:

- Data from a workstation first reaches the central switch
- The switch forward data to the intended destination.
- The router manages internet communication and network security.
- DHCP automatically assigns IP address to devices.

2. Bank network Design using mesh topology

Introduction:

Banks require near-zero downtime making mesh topology the best choice.

Devices Used:

- Core Routers
- Core switches
- Firewalls
- Application servers
- Database servers
- Backup servers
- UPS systems

Transmission media:

- Single-mode fiber optic cable
- Multi-mode fiber optic cable

Network Design:

- Every core network device is connected to multiple devices
- Multiple communication paths exist between nodes
- Redundant internet links are provided through dual routers.

Working Principle:

- Data can be travel through multiple paths.
- If one path fails, traffic automatically switches to another path.
- Routing Protocols such as OSPF and BGP determine the best route.
- Backup power systems maintain operation during power failures.

3. Classroom Laboratory Network Using Bus Topology

Introduction:

Educational institutions often require low-cost networking solutions for computer laboratories. Bus topology offers an economical option by using a single shared communication channel.

Network components:

- computers
- RG-58 coaxial cable
- BNC connectors
- T-connectors
- Terminators
- optional unmanaged switch

Transmission media:

- coaxial cable
- cat5e / cat6 cable

Network Design:

- All computers are connected to a common backbone cable
- Data travels through the shared communication line.
- Terminators are placed at both ends to prevent signal reflection.

Working principle:

- A computer sends data through the backbone cable.
- All devices receive the signal
- Only the intended device processes the data.
- Other devices ignore the transmission.

Conclusion:

Bus topology is suitable for small classroom laboratories where budget is the primary concern.

4. University campus network using Hybrid Topology:

Introduction:

A university campus requires a hybrid topology because it connects multiple buildings and thousands of users.

Network components:

- core routers
- multilayer switches
- Distribution switches
- Access switches
- Wireless access points.

Transmission media:

- Fiber optic cables
- Cat6 cables.

Network Design:

campus Backbone:

- uses Ring or partial mesh topology.
- connects all campus buildings through fiber optics.

Building Level:

- uses star topology
- Distribution switches connect to floor switches.

Laboratory Level:

uses tree or Bus structures for cost efficiency.

Working:

- Fiber backbone connects campus buildings
- star topology is used inside buildings
- Wireless access points provide Wi-Fi access.

company network Design with multiple Departments

Introduction

A company network must support communication between departments while maintaining security and performance

Network components:

- Layer 2 switches
- Layer 3 switch / Router
- Firewall
- Servers
- Workstations
- Printers.

Technologies Used:

- VLANs
- Inter-VLAN Routing
- IP Addressing and subnetting

Network Design:

Access Layer:

Layer 2 switches connect employee devices.

Core Layer:

Layer 3 switch provide connects all departments.

Internet Layer:

Router and Firewall provide internet connectivity and security.

Working Principle:

- Each department has its own VLAN.

- Layer 3 switch / router enables communication between departments.

- Firewall secures internet access.

R. J. S.